Name:- Valle Ramesh

Summary of master class :-

(1.) Data Warehouse :- A centralized repository that stores structured, processed, and organized data for analysis and reporting, mainly used for BI, dashboards, reporting, and business analytics.

(2.) Data Lake :- A storage system that stores raw structured, semi-structured, and unstructured data.

(3.) Data Lakehouse:- Combines the flexibility of a Data Lake with the structure and analytical capabilities of a Data Warehouse.

Medallion Architecture

A **Medallion Architecture** is a data engineering approach that organizes data into **three layers: Bronze, Silver, and Gold**

Bronze — Raw Data

- Stores **raw data exactly as received** from source systems.

Silver — Cleaned Data

- Handles missing values, duplicates, incorrect formats, and standardization.

Gold — Business-Ready Data

- Contains **aggregated and business-focused data** ready for consumption.

(1. **) Dimension table:-** It holds descriptive details. Columns can be primary key.
(2.) **Fact table**:- Holds measurable events. Holds multiple foreign keys, relating to each other.

SCD — Slowly Changing Dimensions

SCD (Slowly Changing Dimension) is a data warehouse technique used to **manage changes in dimension data over time**.

Type 0 - Keep the **original value permanently**; don't update it.

Type 1 - **Overwrite** the old value with the new value; no history maintained.

Type 2 - Create a **new record** for every change; complete history maintained.

Type 3 - Keep **current and previous value** in separate columns; limited history.

Without Generic Script

- SCD logic is written **specifically for one particular table/dataset**.
- The code contains fixed column names, table names, and business rules.

With Generic Script

- SCD logic is written **once and made reusable** for multiple dimension tables.
- Table names, columns, keys, and change-detection rules are passed as **parameters/configuration**.

SSOT vs SVOT

SSOT — Single Source of Truth

- A **single, trusted source** from which consistent and reliable data is obtained.
- Example: One centralized customer database used by Sales, Finance, and Marketing.

SVOT — Single Version of Truth

- Ensures everyone uses the **same, consistent version of data at a particular point in time**.
- Example: All departments see the same **2026 sales figures** from the latest approved dataset.